PAKISTAN NAVY ENGINEERING COLLEGE – NUST

OHT-1 – October 2026 (Sample Paper)

MS (Artificial Intelligence/ Data Science) **CS-871 Machine Learning**

Name: Reg. No:

Time/Shift: 19:00-20:00 PM **Max. marks:** (10 x 6 = **60**)

Special Instructions: 1. This question paper carries 10 technical and conceptual questions.
2. Attempt all questions. Answer briefly and logically. Rubrics are **problem analysis**, **design capability** and **technical justification**.
3. Answer all the questions on the same sheet.
4. Cheating from your peer will straightaway result in **withdrawal of whole semester.**

Q-1: A grocery chain wants to analyze market-basket data, observes that $P(\text{bread} \mid \text{milk}) = 0.7$, and wants to build a product recommendation system. State whether this is best framed as supervised, unsupervised, or association analysis.

Q-2: An NLP startup wants to classify customer support tickets into categories, has raw text data, and wants to convert words into vectors. Explain what faithful embedding is? And why is one-hot embedding often unsuitable for a this large vocabulary-based task.

Q-3: A self-driving car company trains a vision model on daylight images but must deploy it in foggy, low-light conditions, so the test distribution is different from the training distribution. what type of learning is required when the test distribution differs from training.

Q-4: An AI team is building a Mars rover that must navigate autonomously, human expertise does not exist for this environment, and the rover must learn from rewards after each action. Define the learning task T, performance metric P, and experience E, state whether this is supervised, unsupervised, or reinforcement learning

Q-5: A retail company splits its monthly sales data randomly into 80% training and 20% testing; the model performs very well on the test set but fails badly when deployed on next month's sales. State what assumption about the data is violated here, explain why this is a problem for generalization, and describe what validation strategy you would use instead and why.

Q-6: A ride-sharing company predicts trip duration from GPS and traffic data, and a few GPS readings are severe outliers due to sensor glitches; squared loss gives a model that fits these outliers poorly for normal trips. Why squared loss is sensitive to outliers, state which robust loss you would recommend? and how robust loss behaves differently from squared loss for large errors.

Q-7: An e-commerce company has 200 features for purchase prediction, but only a few are actually relevant; a data scientist uses ridge regression and finds that all 200 coefficients remain non-zero. Explain why ridge does not produce sparsity, state which regularizer should be tried instead.

Q-8: A food delivery company wants to predict delivery time, first fits a constant predictor to the delivery times, and the manager says that late deliveries are much worse than early deliveries. State what constant predictor results from squared loss, absolute loss, and which loss you would use if late deliveries are to be penalized more.

Q-9: A robotics team predicts joint angles from sensor readings, and some sensor readings are outliers due to electromagnetic interference, so squared loss produces a poor fit. Explain why robust penalty functions are useful here and why linear or logarithmic growth for large errors helps in the presence of outliers.

Q-10: A hospital uses a 1-nearest neighbor classifier to predict disease from 10 symptoms; it performs perfectly on training data but poorly on new patients, and the features have very different scales, such as age in years and fever in Celsius. Why $k = 1$ may overfit?